

CSE470: Software Engineering

Software Architecture



What is Software Architecture

Architectural design is concerned with understanding **how a software system should be organized and designing** the overall structure of that system.

Software Architecture is how the defining components of a software system are organized and assembled. How they communicate each other. And how the constraints of the whole system is ruled by.

The architectural model serves as an input to the development phase.



Advantages of explicit architecture

Stakeholder communication

- Architecture may be used as a focus of discussion by system stakeholders.

System analysis

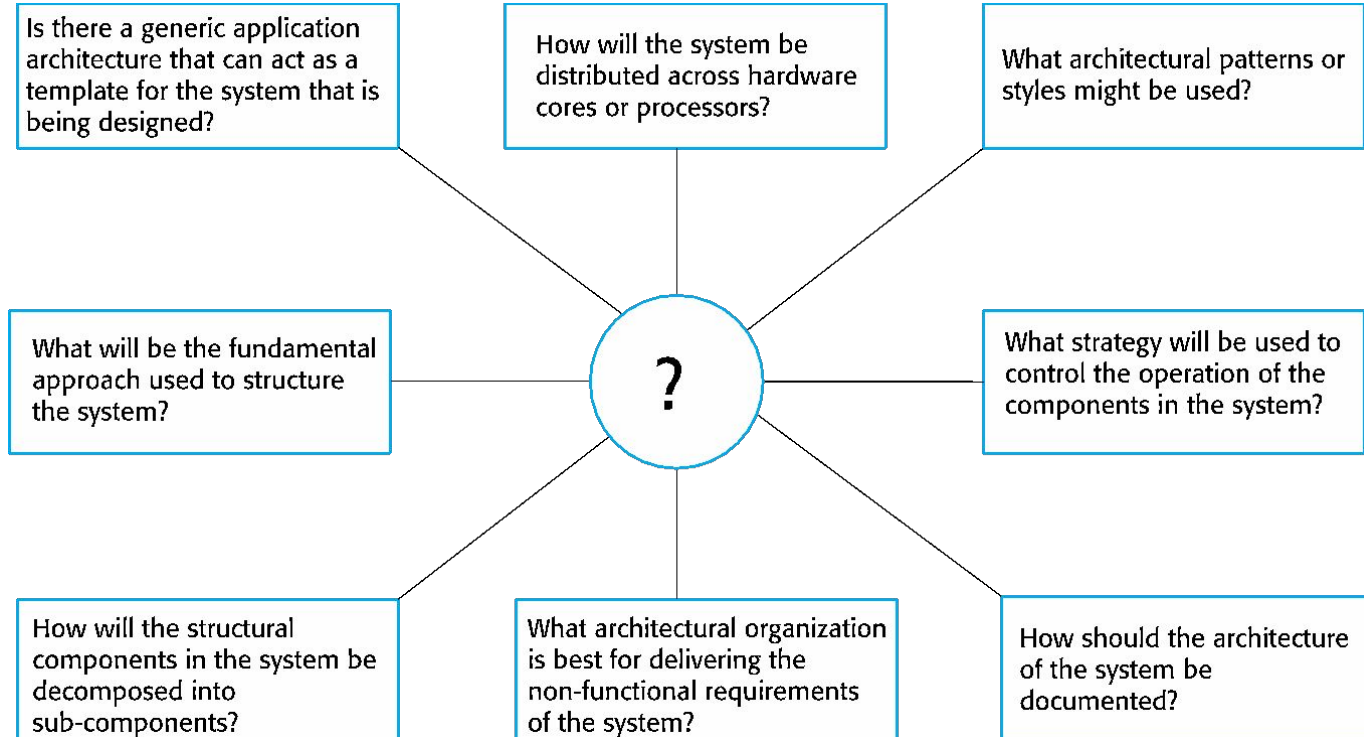
- Means that analysis of whether the system can meet its non-functional requirements is possible.

Large-scale reuse

- The architecture may be reusable across a range of systems
- Product-line architectures may be developed. (Softwares with common features like e-commerce)

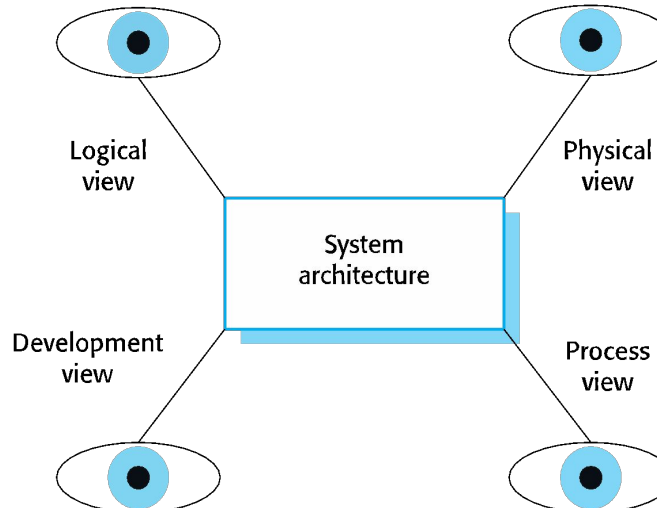


Architecture Design Decisions



Architectural View

- There are four views from which the architecture of a software can be observed
- Each architectural diagram only shows one view or perspective of the system.



Architectural View

View Type	Focus	Example
Logical View	Functional requirements	Class Diagrams, Object Diagrams, State Diagrams.
Development View	Software organization	Component Diagrams, Package Diagrams.
Process View	Dynamic behavior and runtime	Activity Diagrams, Sequence Diagrams.
Physical View	Hardware and topology	Deployment Diagrams.

